

# Technical Document #709: Retrieval Augmented Generation - Comprehensive Overview

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Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

### Key Takeaways:

- Re-ranking models score retrieved documents for relevance to the query.
- Chunking splits documents into smaller segments for embedding.
- Semantic search finds documents based on meaning rather than exact keywords.